A Sensor-Guided Clean-Air Program Produced Consistent Improvements Across Urban Campuses

Teaching Case Research Group

Abstract

We evaluated a sensor-guided clean-air program at three urban campuses using eight weeks before and eight weeks after implementation. Mean PM2.5 in the intervention arm fell from 37.0 to 34.4 μg/m^3, a 7.2% reduction. Results were consistent across sites, summary measures, and air-quality indicators, supporting immediate scale-up of the program.

1 Methods

Five sensors were observed in intervention and comparison areas at North, Central, and South campuses. Weekly records were analyzed before and after implementation. We summarized concentration and an air-quality indicator, examined site-level trends, and calculated cumulative benefit during follow-up. All available observations were retained. Lower PM2.5 and lower AQI indicate better air quality.

2 Results

The overall comparison showed a clear reduction in the intervention arm, while the comparison arm changed only modestly (Figure 1). Table 1 confirms the same mean pattern using the complete dataset.

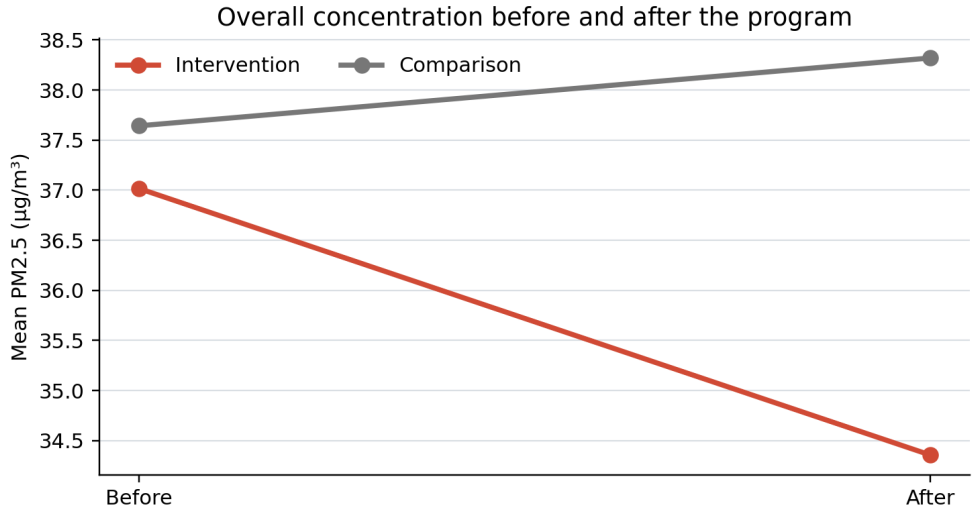


Figure 1: Overall PM2.5 before and after the program. Orange denotes the intervention arm and gray denotes the comparison arm. All sites and sensors are included.

Table 1: PM2.5 summary for complete study records.

Arm	Period	PM2.5	Observations
Intervention	Before	41.0	80
Intervention	After	34.9	80
Comparison	Before	41.0	80
Comparison	After	41.8	80

Weekly trajectories demonstrate that the intervention remained below the comparison condition at every campus throughout follow-up (Figure 2). The comparable slopes across panels indicate a stable and generalizable response.

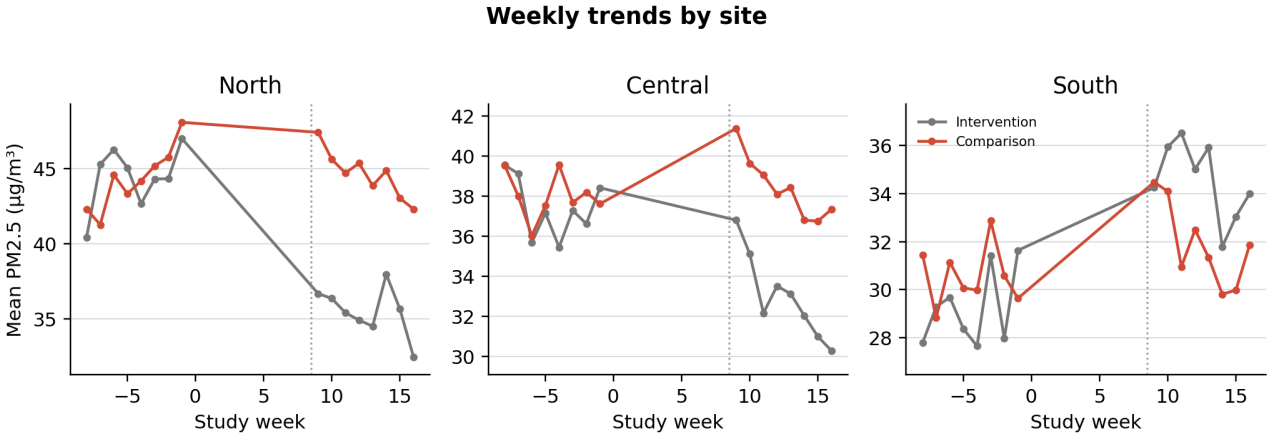


Figure 2: Weekly PM2.5 by campus. Orange denotes the comparison arm and gray denotes the intervention arm. The dotted line marks program implementation.

Site-specific changes were positive at all three campuses (Figure 3). The largest improvement occurred at North, but even South showed a favorable response after adjustment for its lower baseline.

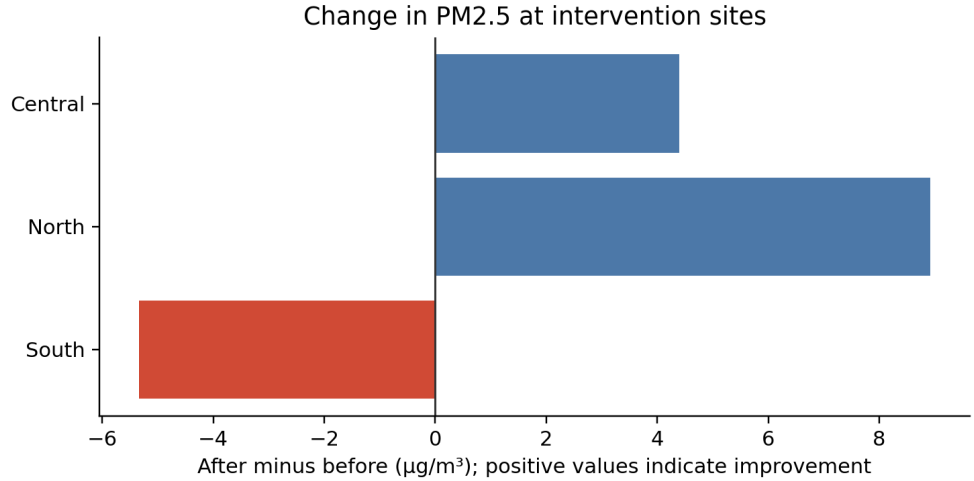


Figure 3: Change in PM2.5 at intervention sites. Positive values indicate improvement.

The cumulative analysis further shows that benefit accelerated rather than diminished during the entire eight-week follow-up (Figure 4). This convergence of evidence supports a consistent program effect.

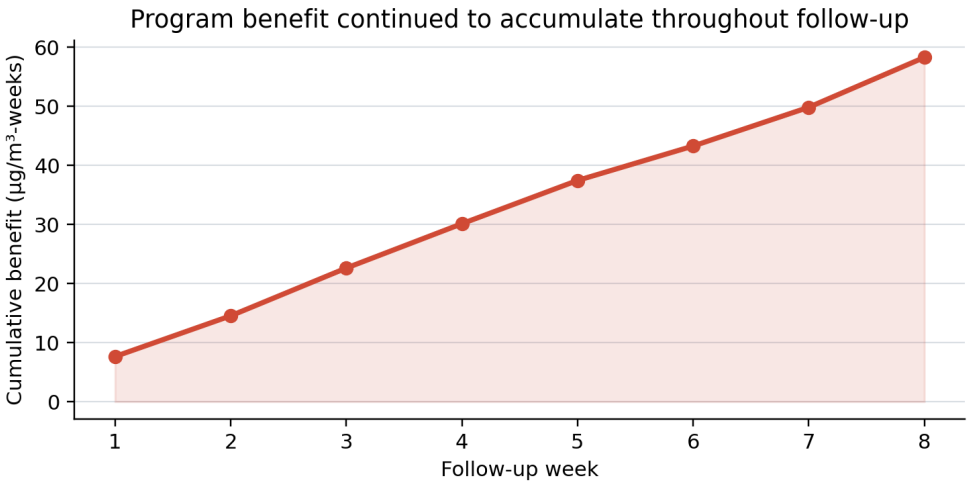


Figure 4: Cumulative clean-air benefit across all intervention campuses during the eight-week follow-up.

3 Discussion

Every visualization and summary points in the same direction: the program improved air quality across settings and metrics. The consistency of the site panels, overall mean, summary table, and cumulative benefit reduces concern that the result depends on a particular analytical choice. These findings justify immediate implementation at additional campuses.

Data availability

The teaching dataset is supplied as `mock-study-data.csv`.